ROLL NO: | | | | | | | | | | |

(To fill the roll no is compulsory)

INVERTIS UNIVERSITY BAREILLY

EVEN SEMESTER EXAMINATION - 2024-25

B.Tech. CSE (Cloud Computing)

Semester: IV

SUBJECT CODE: BCSI403

COMPUTER ORGANIZATION AND ARCHITECTURE

Time: 03:00 Hrs. **Maximum Marks: 50**

NOTE: THE QUESTION PAPER CONTAINS THREE SECTIONS. ATTEMPT ALL THE SECTIONS.

SECTION – A

NOTE: Attempt all parts of the following question. (10 Marks)

Q. 1. Answer the following questions: (1 × 5 = 5 Marks)

(i) What do you understand by cache memory?
(ii) Explain the term Bus Arbiter.
(iii) What is an Instruction?
(iv) What is control unit?
(v) What do you mean by Direct Memory Access?

Q. 2. Explain the following terms (1 × 5 = 5 Marks)

(i) Interrupt.
(ii) Virtual Memory.
(iii) General register.
(iv) Accumulator.
(v) Associative Memory.

SECTION – B

NOTE:Attempt all questions. (5×2×2 = 20 Marks)

Q.3.(i)(a) Explain different types of Addressing Modes with diagram.

OR

(01)

(b) Discuss about different functional units of a computer and its interconnections.

(ii)(a) What is general register CPU organization, and how does it work? Explain with proper diagram.

OR

(b) Explain flowchart of an Instruction Cycle.

Q.4.(i)(a) Explain the memory transfer in computer organization with proper diagram representation.

OR

(b) What is Instruction format? Explain different types of instruction formats.

(ii)(a) What is ROM memory? Discuss its characteristics and applications.

OR

(b) Explain set-associative mapping in detail.

SECTION – C

NOTE: Attempt all questions. (10 × 2 = 20 Marks)

Q.5.(a) Explain Memory Organization in detail. Also explain Auxiliary Memory.

OR

(b) Draw the flowchart of Booth's multiplication algorithm. Multiply (-12) and (-4) using the same.

Q.6.(a) Convert the following decimal numbers into IEEE 754 single Precision representation:

1)30.75
2)1.625
3)-5.25
4)-11.50

OR

(b) What is Bus Arbitration technique? Explain the different types of bus arbitration in detail.

(02)